Li 2019 Melanoma — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Mixed + continuous. Harmony θ-sweep read on the embedding + per-neighbourhood iLISI (1.00→2.00 of 3 samples at θ=5); all 3 samples populate both density lobes (no patient archipelago). One connected component; lobes bridged. Sub-region biology settled: lobes near-equal cycling (0.58/0.64) and no developmental module tracks proliferation → one developmental axis, whole manifold kept.

Gate 2 — module geography → GO (non-branching)

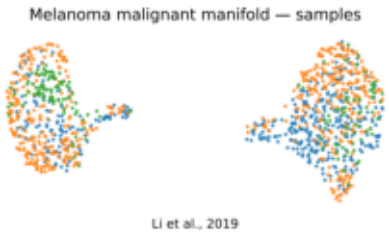
PPT tree over-branches within each lobe at every setting (13–20 forks) = noise-fitting, not a hub→two-arms bifurcation; the manifold is one elongated axis, fit by a fold-free curve (2 tips, 0 forks). Single Gast/pSom-high origin → pSom-leaning axis → inverse terminal Crest gradient (Crest-high decile pseudotime = 1.00). Crest is terminal, not on a Gast arm → no inversion flag.

Gate 3 — hub cross-repression → NA

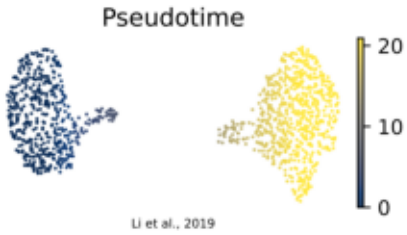
Branching-only. The manifold is non-branching (single continuous curve), so there is no bifurcation hub to test for Gast↔pSom cross-repression. Gate 3 not applicable.

STEP FIGURES

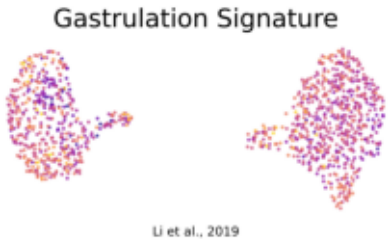
Manifold — samples



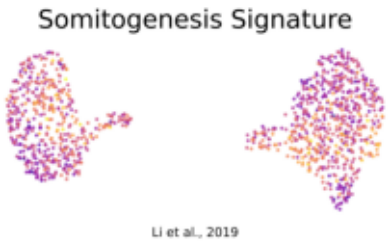
Pseudotime



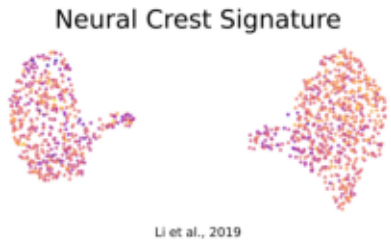
Gastrulation



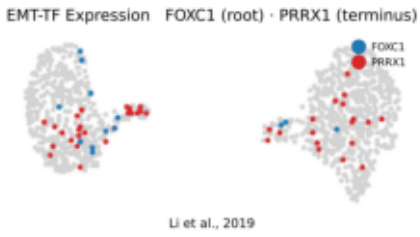
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Melanoma, MARS-seq (Li et al., 2019) — 1,160 malignant cells → 1,082 after per-sample threshold (3 samples).

Verdict.
Pattern reproduces as the non-branching case: one embryonic developmental axis from a Gast/pSom-high origin to a Crest-high terminus, sample-integrated and continuous. Gate 1 GO / Gate 2 GO / Gate 3 NA. Geometry Linear-pSom/Crest; global means Gast −0.015 / pSom +0.016 / Crest −0.002. EMT-TFs FOXC1 (early/root) → PRRX1 (late/terminus).

Parameter choices

Threshold	56 (gap-finding: 154→21, 7.3× drop; derived)	**Integration**	Harmony θ=5 (near-max iLISI mixing; θ=10 fragments the layout); n_pcs=20
Root	Gast/pSom-high curve tip (origin; Crest late) — not a proliferation state	**Geometry**	Non-branching single continuous curve (Linear-pSom/Crest); means Gast −0.015 / pSom +0.016 / Crest −0.002
Cell cycle	scored (no publication phase calls); no module tracks proliferation	**EMT-TFs**	FOXC1 (early/root) → PRRX1 (late/terminus), detection-gated